Florian **Franzen**

Leadership | Architecture | Infrastructure | Research

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Technical leader for complex infrastructure and protocol systems, from specification through implementation, deployment and operations. From setting technical strategy, research and roadmaps and making the architecture decisions to hiring for the team and driving partnerships and integrations, I will take systems from idea to production and keep them running there. I have led engineering, research and infrastructure for Substrate-, Ethereum- and Solana-based networks through to mainnet launch. Today I provide technical leadership and engineering support to startups building protocols and infrastructure, through my studio F.R.E.D.

Behind that sits an unusually deep range: years in neuroscience and academic engineering, embedded hardware and prototype manufacturing, security and reverse engineering, data analysis and machine learning (including AI/LLMs), and full-stack web development. It lets me own a system from the circuit board to the production network.

Skills

Leadership	technical strategy, roadmaps, architecture decisions, hiring, partnerships and integrations, due diligence
Languages	**Rust** (expert), **C/C++** (expert), Go (advanced), **Python** (advanced), TypeScript (proficient), Haskell (proficient), SQL
Protocols	Polkadot (**Substrate**, PolkaVM), Solana (**Anchor**), Ethereum (**Solidity**, Hardhat)
Infrastructure	Ansible, Docker/Podman, Kubernetes/Helm, **Nix/NixOS**, CI/CD, **Networking**, **Linux**
Tooling	**Git**, **CMake**, Ghidra, IDA Pro, **Fuzzing**, AsciiDoc, Doxygen, LaTeX
Hardware	embedded development, PCB design (KiCad), 3D printing/CAD, electronics repair
Spoken Languages	English (fluent), German (native)

Professional Experience

Current – Jun. 2026 — Head of Research, Flowra, Remote
Partner with the CTO on the block engine: monitor performance, investigate deviations, review code changes and propose alternative approaches. Track competing block engines and developments across the Solana network. Publish regular articles on protocol changes and their impact on product strategy.
Agave · Alpenglow · Firedancer · Jito · MEV · Rust · Solana · TypeScript · Tokio

Apr. 2026 – Dec. 2025 — Senior Research Engineer, Raiku Labs Ltd, Remote
Design and implement auction marketplace. Interface with on-chain programs and data. Optimize performance, scheduling and blockspace utilization. Support DevOps with upcoming changes. Liaison for the Research team.
Anchor · Helm · Jito · Kubernetes · MEV · Rust · Solana · Tokio · Tonic

Dec. 2025 – Aug. 2025 — Head of Protocol Research, Gluwa, Remote
Direct research strategy and execution. Support product goals across technical teams. Lead evaluation of emerging technologies and advise the Chief Architect and Head of Engineering on critical technical decisions.
Cairo · Python · Rust · Solidity · Substrate · TypeScript

Aug. 2025 – Dec. 2024 — Head of Infrastructure, Analog, Remote
Oversee core blockchain infrastructure: mainnet launch, feature development, DevOps/SecOps, exchange and wallet integrations. Execute on-chain governance, fund management and runtime upgrades alongside founders.
Alloy · GraphQL · Hardhat · Helm · Kubernetes · PolkaVM · React · Rust · Substrate · TypeScript · Tokenomics · WASM

Dec. 2024 – Jun. 2023 — Head of Research, Analog, Remote
Research and prototype core protocol components. Lead DevOps/SecOps and the technical side of product management, hiring and partner integrations. Report directly to founders.
Cryptography · Game Theory · Helm · Kubernetes · Protocol Design · Rust · Security · Substrate · Tokenomics · WASM

May. 2023 – Mar. 2020 — Polkadot Specification Lead, Web3 Foundation, Zug, Switzerland
Co-develop the implementable Polkadot specification alongside the research team. Validate external implementations through a custom cross-implementation test suite. Shepherd the multi-client standardization process.
AsciiDoc · C++ · CMake · Go · Gossamer · Kagome · Julia · k-framework · Python · Ruby · Rust · Substrate · WASM

May. 2019 – Jul. 2016 — Doctoral Researcher in Neuroinformatics, Research Center caesar, Bonn, Germany
Develop 3-D environment and pose reconstruction from video data using non-linear optimization of multi-view geometric graphs. Build an ultra-lightweight tetherless recording platform for GPS, IMU, and barometric sensors. Reverse-engineer and extend a consumer-grade 360° camera. Develop advanced camera calibration, synchronization, and spatial tracking system.
Blender · C · C++ · CMake · Docker · IDA Pro · i²c · Jupyter · Matlab · Nix · OpenCV · Python · LSD-SLAM · SPI · Teensy · Qt

Jul. 2016 – Nov. 2015 | **Network & System Administrator, Research Center caesar, Bonn, Germany**
Install and configure wired and wireless infrastructure for 200+ clients, including firewalls, routing, switching, VLANs, 802.1x authentication, VPN, RADIUS, and SIP telephony.
Bash | Cisco ASA | Cisco IOS | Cisco ISE | Cisco WLC | Microsoft Active Directory

Freelance Experience

Current – 2023 | **Founder & Technical Lead, Franzen Research, Engineering & Development**
Technical leadership and engineering support for startups building protocols and infrastructure: architecture, protocol research, infrastructure and validator operations, security. Prototyping and embedded development, scientific computing, data digitization and electronics repair.
C/C++ | CXADC | ESP32 | KiCad | Nix | Ollama | OpenSCAD | ROCm | Rust | Smartcard | VHS-Decode

Current | **Integration Consultant, Stealth Startup**
Advise on protocol architecture, mainnet integration strategy, and genesis launch execution.
Intents | MEV-protection | Quantum-safety | Rust | Sharding

2025 | **Product and DevOps Engineer, Analog**
Design, implement, and deploy the official Substrate-Ethereum bridge. Containerize infrastructure for Kubernetes/Helm deployments, optimize reliability and cost, manage on-chain runtime upgrades and other governance.
Ansible | Docker | Helm | Kubernetes | Linux | Podman | React | Rust | Solidity | Substrate | TypeScript | WASM

2019 – 2018 | **Software and Network Engineer, White Matter LLC**
Port a C++ API from Winsock to cross-platform Asio. Design binary network protocols, optimize packet throughput and stability, enforce thread safety, implement unit tests, and extend the OpenEphys plugin.
Asio | C++17 | Catch2 | CMake | Doxygen | fmtlib | JUCE | Linux | spdlog | Windows

Education

2014 | **Master of Neural & Behavioural Science, International Max Planck Research School, Tübingen, Germany**
Thesis: "Evaluation of a Statistical Model for Cortical State Identification"
Final grade: 2.1

2012 | **Bachelor of Cognitive Science, University of Tübingen, Tübingen, Germany**
Thesis: "Neural Circuits of Locomotor-related Response Modulation in Mouse Primary Visual Cortex"
Final grade: 2.1

Publications

2014 P. Putzky, **F. Franzen**, G. Bassetto, J. H. Macke. A Bayesian model for identifying hierarchically organised states in neural population activity. *Advances in Neural Information Processing Systems 27*. 2014.

2013 A. Vaiceliunaite, S. Erisken, **F. Franzen**, S. Katzner, L. Busse. Spatial integration in mouse primary visual cortex. *J Neurophysiol*. 2013 May 29.

Teaching

2023 – *10 days* | **Polkadot Blockchain Academy, Universidad de Buenos Aires, Buenos Aires, Argentina**
Teaching cryptography, economics and blockchain

2022 – *1 week* | **Polkadot Blockchain Academy, Corpus Christi College, Cambridge, England**
Teaching economics through interactive games

2015 – *2 weeks* | **Open Lab Ware Course, TReND in Africa, Addis Ababa, Ethiopia**
Building and calibrating delta-style 3D printers, CAD, electrical circuit design, microcontroller programming